UNIVERSITY OF ENGINEERING & TECHNOLOGY MARDAN

COMPUTER SCIENCE

CS-305L: Parallel & Distributed Computing Lab

SUBMITTED BY

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REG NO : 23MDBCS 435

SECTION : CS-A

SEMESTER : 6th

LAB 4

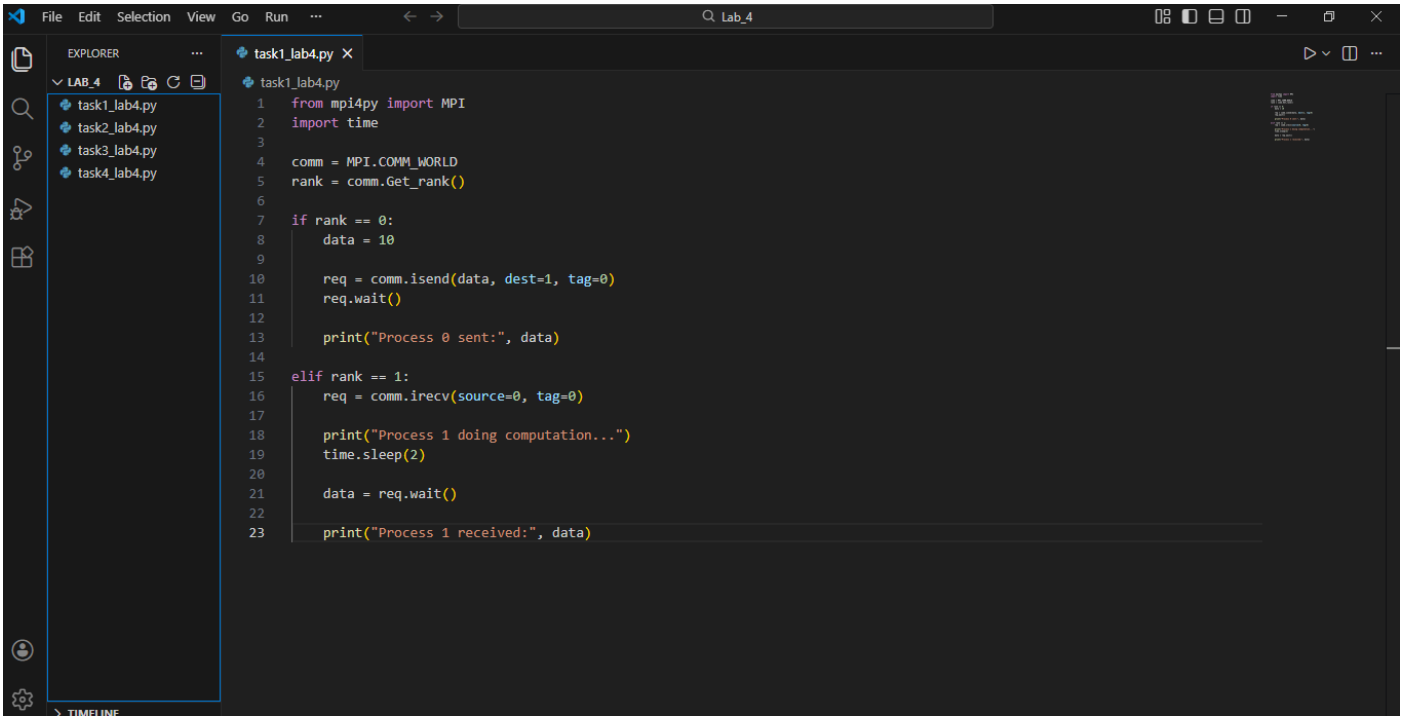
Task 1: Non-blocking Communication (Isend/Irecv)

• Implement an MPI program where **Process 0** sends an integer to **Process 1** using **nonblocking communication**(Isend and Irecv).

• Ensure that the receiving process performs a computation before printing the received value.

• How does Isend differ from send, and why is Wait() necessary in non-blocking communication?

Program



Output

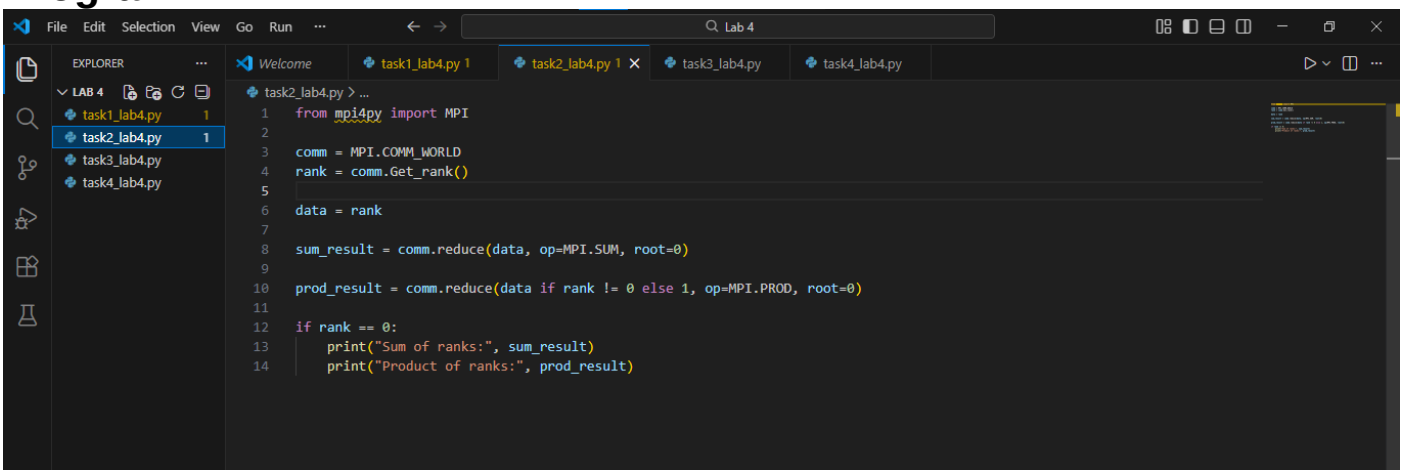
```
(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_4>mpiexec -n 2 python task1_lab4.py
Process 0 sent: 10
Process 1 doing computation...
Process 1 received: 10

(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_4>
```

Task 2: Reduce Operation

• Write an MPI program where each process initializes a variable with its **rank number** and uses the **Reduce operation** to compute the sum of all ranks at **Process 0**.

• Modify the program to use MPI.PROD instead of MPI.SUM. What difference do you observe in the result?

Program



Output

```
(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_4>mpiexec -n 4 python task2_lab4.py
Sum of ranks: 6
Product of ranks: 6

(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_4>
```

Task 3: Gather Operation (MPI.Gather)

• Implement an MPI program where each process initializes a number **equal to its rank** and sends it to **Process 0**using MPI.Gather.

• What happens if recv_data is not initialized correctly at the root process?

• Modify the program to use MPI.Gatherv for gathering different amounts of data from each process.

Program

Ouput

```
(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_4>mpiexec -n 4 python task3_lab4.py
Gathered Data: [0 1 2 3]

(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_4>
```

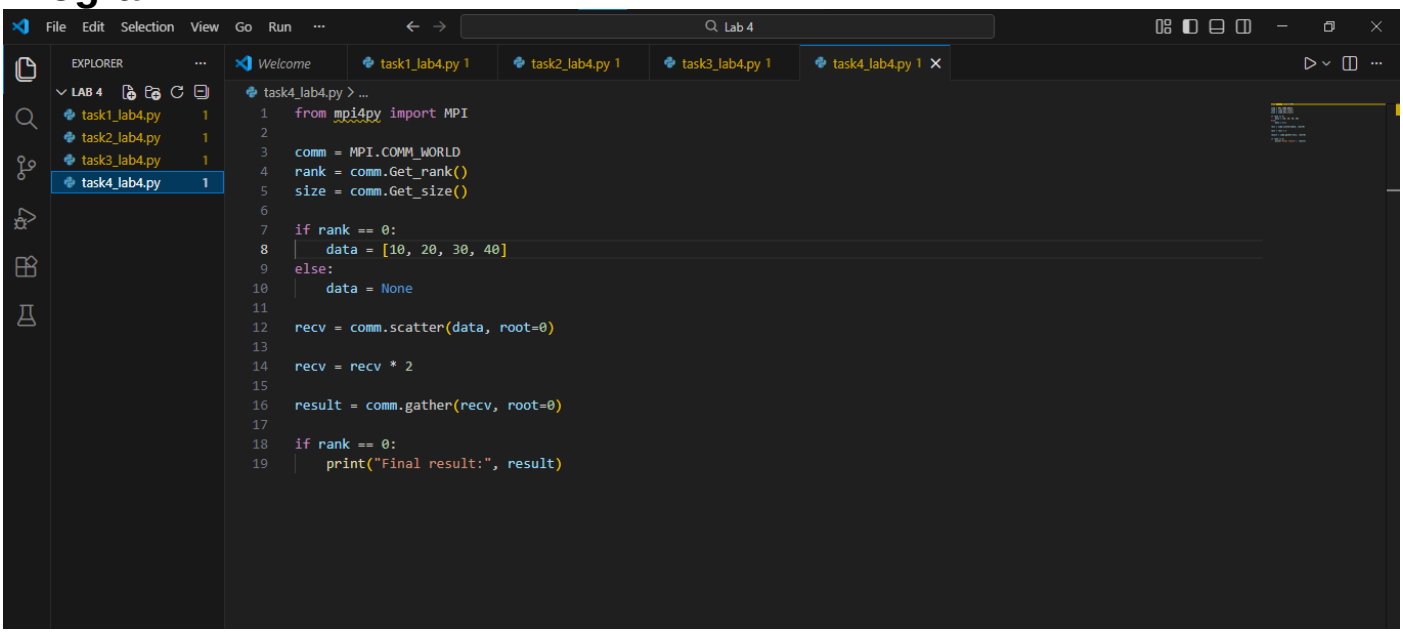
Task 4: Scatter Operation (MPI.Scatter)

• Write an MPI program where **Process 0** initializes an array [10, 20, 30, 40] (for 4 processes) and distributes one value to each process using MPI.Scatter.

• What happens if the number of processes is different from the number of elements in the array?

• Modify the program so that after receiving data, each process **doubles** its value and sends it back to **Process 0** using MPI.Gather.

Program



Ouput

```
(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_4>mpiexec -n 4 python task4_lab4.py
Final result: [20, 40, 60, 80]

(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_4>
```